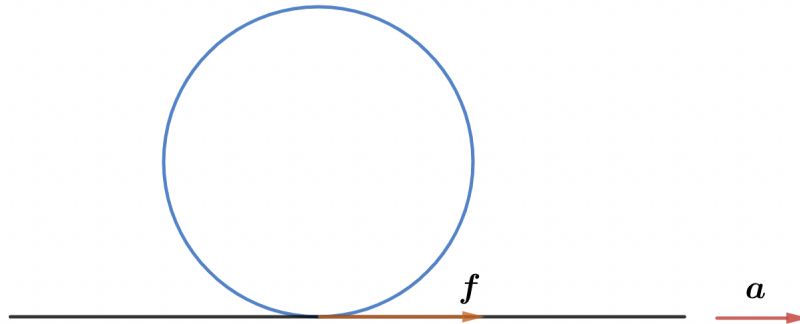
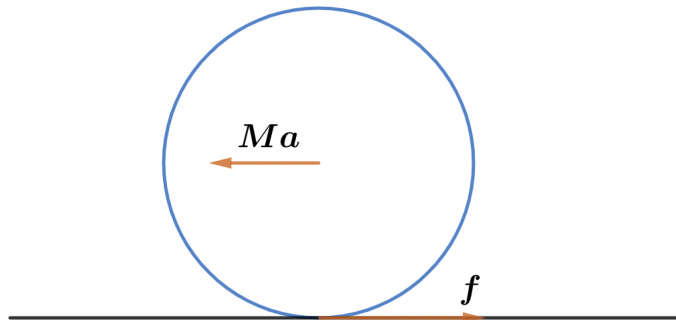


2022B F=ma Exam: Problem 20

Kevin S. Huang



We can simplify the problem by going to the accelerating frame and adding the fictitious force on the sphere:



Letting leftwards be the positive direction, Newton's 2nd law yields

$$Ma - f = Ma_t$$

$$fR = I\alpha$$

Since the sphere doesn't slip, we have

$$\alpha = \frac{a_t}{R}$$

Combining the first two equations,

$$Ma = Ma_t + \frac{Ia_t}{R^2} = a_t \left(M + \frac{2}{3}M \right) = \frac{5Ma_t}{3}$$

using the moment of inertia of a shell $I = \frac{2}{3}MR^2$. Thus,

$$a_t = \frac{3}{5}a$$

We have to add acceleration a back to return to the ground frame:

$$a_t^{\text{ground}} = a_t - a = -\frac{2}{5}a$$

By our sign convention, the answer is D.